

THE HERITAGE SCHOOL
FIRST TERM EXAMINATION, 2026-27
PHYSICS (THEORY) · CLASS 12 · UNIT TEST 1

Maximum Marks: 40

Time Allotted: One and a Half Hours

Instructions to Candidates

- You are allowed an additional fifteen minutes for only reading the question paper. You must NOT start writing during this time.
- There are **twelve questions** in this paper. **Answer all questions.**
- There are **four sections** in the paper: A, B, C and D. Internal choices have been provided in **two** questions in Section B, **one** question in Section C and **two** questions in Section D.
- **Section A** consists of one question having eight sub-parts of **1 mark** each.
- While attempting the multiple choice sub-parts you are required to write only ONE option as the answer.
- **Section B** consists of five questions of **2 marks** each.
- **Section C** consists of four questions of **3 marks** each.
- **Section D** consists of two questions of **5 marks** each.
- The intended marks for questions or parts of questions are given in brackets [].
- All working, including rough work, should be done on the same sheet as, and adjacent to, the rest of the answer.
- A list of useful physical constants is given at the end of this paper.
- A simple scientific calculator without a programmable memory may be used for calculations.

Syllabus covered: Electrostatics; Current Electricity; Magnetic Effect of Current and Magnetism.

Not examined in this paper: Electromagnetic Induction and Alternating Current, Electromagnetic Waves and Optics are not examined here; they are in the First Term syllabus but outside this test.

SECTION A - 8 MARKS

Question 1 [8 × 1]

In sub-parts (i) to (iv) choose the correct alternative. Answer sub-parts (v) to (viii) briefly.

- (i) Two identical conducting spheres carrying charges $+q$ and $-3q$ are brought into contact and then separated so that their centres are a distance r apart. The force between them is:
- (a) attractive, of magnitude kq^2/r^2 .
 - (b) repulsive, of magnitude kq^2/r^2 .
 - (c) repulsive, of magnitude $3kq^2/r^2$.
 - (d) zero.
- (ii) A wire of resistance R is cut into four equal parts, which are then joined in parallel. The resistance of the combination is:
- (a) $R/4$
 - (b) $R/8$
 - (c) $R/16$
 - (d) $4R$
- (iii) The magnetic flux density well inside a long current-carrying solenoid is B . At a point on its axis at one **end** of the solenoid it is:
- (a) B
 - (b) $B/2$
 - (c) $2B$
 - (d) zero

- (iv) **Assertion:** a current-carrying solenoid behaves like a bar magnet. **Reason:** the magnetic field lines outside a finite solenoid are of the same form as those of a bar magnet, leaving one face and entering the other.
- (a) Both Assertion and Reason are true and Reason is the correct explanation for Assertion.
(b) Both Assertion and Reason are true but Reason is not the correct explanation for Assertion.
(c) Assertion is true and Reason is false.
(d) Assertion is false and Reason is true.
- (v) Why must the electrostatic field at the surface of a charged conductor be normal to the surface at every point?
- (vi) A dielectric slab is introduced between the plates of an **isolated** charged parallel plate capacitor. State what happens to the potential difference across it.
- (vii) Why is manganin preferred to copper for making standard resistance coils?
- (viii) State how the resistance of a semiconductor changes when its temperature is raised, and give the reason.

SECTION B - 10 MARKS

Question 2 [2]

Capacitors of $4.0\ \mu\text{F}$ and $12\ \mu\text{F}$ are connected in series across a $100\ \text{V}$ supply. Calculate the charge on each capacitor and the potential difference across the $4.0\ \mu\text{F}$ capacitor.

Question 3 [2]

A cell of emf $1.5\ \text{V}$ and internal resistance $0.50\ \Omega$ is short-circuited. Calculate the current that flows and state the terminal potential difference.

OR

State any two factors on which the internal resistance of a cell depends.

Question 4 [2]

Resistors of $4.0\ \Omega$ and $6.0\ \Omega$ are joined in series across a $20\ \text{V}$ supply of negligible internal resistance. Calculate the potential difference across the $6.0\ \Omega$ resistor.

Question 5 [2]

A moving coil galvanometer of resistance $50\ \Omega$ gives a full-scale deflection for a current of $2.0\ \text{mA}$. How would you convert it into a voltmeter reading up to $10\ \text{V}$?

Question 6 [2]

State Ampère's circuital law.

OR

Two identical circular coils, each of radius R and carrying a current I , are placed concentrically with their planes at right angles to each other. Find the magnetic flux density at their common centre.

SECTION C - 12 MARKS

Question 7 [3]

Using Gauss' theorem, derive an expression for the electric field intensity due to an infinite plane sheet of charge of surface charge density σ . Hence write down the field just outside a charged conductor.

Question 8 [3]

State Kirchhoff's two laws for an electrical network, and use them to obtain the condition of balance of a Wheatstone bridge.

OR

Resistors of $6.0\ \Omega$ and $3.0\ \Omega$ are connected in parallel across a battery of emf $12\ \text{V}$ and internal resistance $1.0\ \Omega$. Calculate the current drawn from the battery and the current in each resistor.

Question 9 [3]

Draw a labelled circuit diagram of a potentiometer used to compare the emfs of two cells, and write the working formula, explaining the symbols. (Derivation is not required.)

Question 10 [3]

Derive an expression for the force per unit length between two long, straight, parallel conductors carrying currents I_1 and I_2 , separated by a distance r . Hence define the ampere.

SECTION D - 10 MARKS**Question 11 [5]**

(a) Derive an expression for the capacitance of a parallel plate capacitor when a dielectric slab of thickness t and dielectric constant K is introduced between its plates, the plate separation being d ($t < d$). [3]

(b) A parallel plate air capacitor has a capacitance of 8.0 pF. Its plate separation is halved and the space between the plates is completely filled with a medium of dielectric constant 5.0. Calculate the new capacitance. [2]

OR

(a) Derive an expression for the potential energy of an electric dipole of moment p placed at an angle θ to a uniform electric field E , and state the positions of stable and unstable equilibrium. [3]

(b) A dipole of moment 2.0×10^{-9} C m is held in a uniform field of 3.0×10^5 N C⁻¹. Calculate the work done in rotating it from the position of stable equilibrium through 180°. [2]

Question 12 [5]

(a) Using Ampère's circuital law, derive an expression for the magnetic flux density inside a long straight solenoid carrying a current I , having n turns per unit length. [3]

(b) A solenoid 50 cm long has 500 turns and carries a current of 2.0 A. Calculate the magnetic flux density near its centre. [2]

OR

(a) Derive an expression for the torque on a rectangular coil of N turns and area A , carrying a current I , placed in a uniform magnetic field B . [3]

(b) A coil of 50 turns and area 2.0×10^{-2} m² carries a current of 3.0 A in a uniform field of 0.10 T, with the **plane of the coil parallel** to the field. Calculate the torque on it. [2]

USEFUL CONSTANTS AND RELATIONS

	Quantity	Symbol	Value
1.	Permittivity of vacuum	ϵ_0	8.85×10^{-12} F m ⁻¹
2.	Constant for Coulomb's law	$1/4\pi\epsilon_0$	9×10^9 N m ² C ⁻²
3.	Permeability of vacuum	μ_0	$4\pi \times 10^{-7}$ H m ⁻¹
4.	Constant in magnetism	$\mu_0/4\pi$	1×10^{-7} H m ⁻¹
5.	Charge of a proton = charge of an electron	e	1.6×10^{-19} C
6.	Mass of an electron	m_e	9.1×10^{-31} kg
7.	Mass of a proton	m_p	1.67×10^{-27} kg
8.	Speed of light in vacuum	c	3×10^8 m s ⁻¹
9.	Acceleration due to gravity	g	9.8 m s ⁻²
10.	Least distance of distinct vision	D	25 cm

ANSWER KEY & MARKING NOTES

Unit Test 1. **Do not read this section before attempting the paper.** Every numerical answer was recomputed independently; the notes give the decisive step rather than a full solution.

Section A

Q	Answer
1 (i)	(b) On contact the charge is shared equally, so each carries $(q - 3q)/2 = -q$. Both are now negative , so the force is repulsive and of magnitude kq^2/r^2 .
1 (ii)	(c) Each part has resistance $R/4$, and four of them in parallel give $(R/4)/4 = \mathbf{R/16}$. In general cutting into n parts and paralleling them divides the resistance by n^2 .
1 (iii)	(b) $B/2$. At the end of a long solenoid only half the winding contributes on either side, so the field falls to half its interior value.
1 (iv)	(a) Both statements are true, and the resemblance of the external field is precisely why the solenoid has a north and a south face.
1 (v)	Because a tangential component would exert a force on the free surface charges and set up a current along the surface. In electrostatic equilibrium no such current flows, so the tangential component must be zero and the field is entirely normal.
1 (vi)	It decreases , by the factor K . The charge is fixed (the capacitor is isolated) while C rises to KC , so $V = Q/C$ falls to V/K .
1 (vii)	Because manganin has a very low temperature coefficient of resistance , so its resistance hardly changes as it warms up in use, and a high resistivity so a useful resistance needs only a short length. Copper's resistance would drift with temperature.
1 (viii)	It decreases . Raising the temperature releases many more charge carriers, so n in $\rho = m/ne^2\tau$ rises far faster than τ falls — the opposite of a metal, where n is fixed and only τ changes.

Section B

Q	Answer
2	$C = (4)(12)/16 = 3.0 \mu\text{F}$, so $Q = CV = (3.0 \times 10^{-6})(100) = \mathbf{3.0 \times 10^{-4} \text{ C} = 300 \mu\text{C}}$, the same on each (series). Then $V_4 = Q/C_4 = 300/4.0 = \mathbf{75 \text{ V}}$ (and 25 V across the 12 μF — the smaller capacitor takes the larger share).
3	With the terminals joined by a conductor of negligible resistance, $I = \mathcal{E}/r = 1.5/0.50 = \mathbf{3.0 \text{ A}}$. The terminal potential difference is $V = \mathcal{E} - Ir = 1.5 - 1.5 = \mathbf{\text{zero}}$: the whole emf is used up driving the current through r .
3 (OR)	Any two of: the separation of the electrodes (larger separation, larger r); the area of the electrodes dipped in the electrolyte (larger area, smaller r); the concentration of the electrolyte ; and the temperature (r falls as the temperature rises).
4	$I = 20/(4.0+6.0) = 2.0 \text{ A}$, the same in both. $V_6 = IR = (2.0)(6.0) = \mathbf{12 \text{ V}}$ (the 4.0 Ω takes the remaining 8.0 V, so the supply voltage divides in the ratio of the resistances).
5	Connect a high resistance in series with the galvanometer, of value $R = V/I_g - G = 10/(2.0 \times 10^{-3}) - 50 = 5000 - 50 = \mathbf{4950 \Omega}$. The voltmeter's resistance is then 5000 Ω , high enough that it draws almost no current from the circuit.
6	The line integral of the magnetic field around any closed loop equals μ_0 times the net current threading that loop: $\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I$. It is useful only where the symmetry lets B be taken out of the integral.
6 (OR)	Each coil alone gives $B_0 = \mu_0 I/2R$ at the centre, and the two fields are along the two axes, i.e. perpendicular to each other. They therefore add vectorially: $B = \sqrt{(B_0^2 + B_0^2)} = \mathbf{\sqrt{2} \mu_0 I/2R}$, at 45° to each axis.

Section C

Q	Answer
7	Take a cylindrical Gaussian surface of cross-section A piercing the sheet symmetrically. By symmetry $\mathbf{E}$ is perpendicular to the sheet and of equal magnitude on the two flat faces; the flux through the curved surface is zero . [1] So $\Phi = 2EA = \sigma A/\epsilon_0$. [1] Hence $\mathbf{E} = \sigma/2\epsilon_0$, independent of the distance from the sheet. For a charged conductor the field exists on one side only, so the same calculation gives $\mathbf{E} = \sigma/\epsilon_0$ — twice as large. [1]
8	Junction rule: the algebraic sum of the currents at a junction is zero (conservation of charge). Loop rule: the algebraic sum of the potential differences around any closed loop is zero (conservation of energy). [1] At balance no current flows through the galvanometer , so I_1 passes through P and Q , and I_2 through R and S . [1] Applying the loop rule to the two halves: $I_1P = I_2R$ and $I_1Q = I_2S$. Dividing, $P/Q = R/S$. [1]
8 (OR)	Parallel combination: $(6)(3)/9 = 2.0 \Omega$; total resistance $2.0 + 1.0 = 3.0 \Omega$. [1] $I = 12/3.0 = 4.0 \text{ A}$ from the battery, and the potential difference across the parallel pair is $(4.0)(2.0) = 8.0 \text{ V}$. [1] Hence $I_6 = 8.0/6.0 = 1.3 \text{ A}$ and $I_3 = 8.0/3.0 = 2.7 \text{ A}$, which sum to 4.0 A as the junction rule requires. [1]
9	Diagram: driver cell with key and rheostat in series with the potentiometer wire AB ; the two cells ϵ_1 and ϵ_2 with their positive terminals joined to A through a two-way key, the other ends going through a galvanometer to the jockey. [2] With balancing lengths l_1 and l_2 for the two cells, each emf equals the potential drop over its balancing length, so $\epsilon_1/\epsilon_2 = l_1/l_2$. [1] The driver emf must exceed both, or no balance point exists on the wire.
10	The first wire produces at the second a field $B_1 = \mu_0 I_1/2\pi r$, perpendicular to it. [1] A length l of the second wire therefore feels $F = B_1 I_2 l$, so $F/l = \mu_0 I_1 I_2/2\pi r$ — attractive for parallel currents and repulsive for antiparallel ones. [1] Definition: one ampere is that steady current which, flowing in each of two infinitely long parallel conductors of negligible cross-section placed 1 m apart in vacuum , produces a force of $2 \times 10^{-7} \text{ N per metre}$ of their length. [1]

Section D

Q	Answer
11	(a) The field is $E_0 = \sigma/\epsilon_0$ over the air gap of thickness $(d-t)$ and is reduced to E_0/K inside the slab. [1] So $V = E_0(d-t) + (E_0/K)t = E_0[(d-t) + t/K]$. [1] With $C = Q/V$ and $E_0 = Q/\epsilon_0 A$, $C = \epsilon_0 A/[d - t(1 - 1/K)]$. [1] Putting $t = d$ recovers $C = K\epsilon_0 A/d$, and $t = 0$ recovers the air value — a useful check. (b) $C = \epsilon_0 A/d = 8.0 \text{ pF}$. Halving d doubles it and filling with $K = 5.0$ multiplies it by five, so $C' = K\epsilon_0 A/(d/2) = 2KC = (2)(5.0)(8.0) = 80 \text{ pF}$.
11 (OR)	(a) The torque needed to hold the dipole at an angle θ is $\tau = pE \sin \theta$, so the work done in turning it through $d\theta$ is $dW = pE \sin \theta d\theta$. [1] Integrating from the reference position $\theta = 90^\circ$, where the energy is taken as zero, $U = \int pE \sin \theta d\theta$ from 90° to θ , giving $U = -pE \cos \theta = -\mathbf{p} \cdot \mathbf{E}$. [1] U is least ($= -pE$) at $\theta = 0$, which is stable equilibrium, and greatest ($= +pE$) at $\theta = 180^\circ$, which is unstable . [1] (b) $W = U_{180^\circ} - U_{0^\circ} = pE - (-pE) = 2pE = 2(2.0 \times 10^{-9})(3.0 \times 10^5) = 1.2 \times 10^{-3} \text{ J}$.
12	(a) Take a rectangular Ampèrian loop with one long side of length L inside the solenoid, parallel to the axis, and the opposite side well outside. [1] Outside a long solenoid the field is negligible, and on the two short sides $\mathbf{B} \perp d\mathbf{l}$, so only the inner side contributes: $\oint \mathbf{B} \cdot d\mathbf{l} = BL$. [1] The current threading the loop is nLI , so $BL = \mu_0 nLI$, giving $\mathbf{B} = \mu_0 n\mathbf{l}$ — independent of position, i.e. the field inside is uniform , and it is $B/2$ at the ends. [1] (b) $n = 500/0.50 = 1000 \text{ turns m}^{-1}$, so $B = (4\pi \times 10^{-7})(1000)(2.0) = 2.5 \times 10^{-3} \text{ T} (= 2.5 \text{ mT})$, directed along the axis.
12 (OR)	(a) The two sides of length l perpendicular to $\mathbf{B}$ each carry a force $F = BIl$. They are equal, opposite and not collinear , so they form a couple; the forces on the other two sides are collinear and cancel. [1] Their perpendicular separation is $b \sin \theta$, where θ is the angle between the normal to the coil and $\mathbf{B}$, so $\tau = BIl \cdot b \sin \theta = BIA \sin \theta$. [1] For N turns $\tau = NIAB \sin \theta$, or vectorially $\tau = \mathbf{m} \times \mathbf{B}$ with $\mathbf{m} = NIA$. [1] (b) The plane is parallel to the field, so the normal is perpendicular to it: $\theta = 90^\circ$ and $\sin \theta = 1$, which is the position of maximum torque. $\tau = (50)(3.0)(2.0 \times 10^{-2})(0.10) = 0.30 \text{ N m}$.

Mark distribution over the chapters examined

Chapter	A	B	C	D	Total
Electrostatics	3	2	3	5	13
Current Electricity	3	4	6	—	13
Magnetic Effect of Current and Magnetism	2	4	3	5	14
Total	8	10	12	10	40